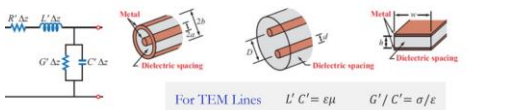
Parameter	Coaxial	Two-Wire	Parallel-Plate	Unit
R'	$\frac{R_s}{2\pi}\left(\frac{1}{a}+\frac{1}{b}\right)$	$\frac{2R_s}{\pi d}$	$\frac{2R_s}{w}$	Ω/m
L'	$\frac{\mu}{2\pi}\ln(b/a)$	$\frac{\mu}{\pi}\ln\left[(D/d)+\sqrt{(D/d)^2-1}\right]$	$\frac{\mu h}{w}$	H/m
G'	$\frac{2\pi\sigma}{\ln(b/a)}$	$\frac{\pi\sigma}{\ln\left[(D/d)+\sqrt{(D/d)^2-1}\right]}$	$\frac{\sigma w}{h}$	S/m
C'	$\frac{2\pi\varepsilon}{\ln(b/a)}$	$\frac{\pi\varepsilon}{\ln\left[(D/d)+\sqrt{(D/d)^2-1}\right]}$	$\frac{\varepsilon w}{h}$	F/m



For TEM Lines $L'C' = \varepsilon\mu$ $\quad G'/C' = \sigma/\varepsilon$

Insert the two equations into each other.

$$\frac{d^2V(z)}{dz^2} - \gamma^2 V(z) = 0$$

$$\frac{d^2I(z)}{dz^2} - \gamma^2 I(z) = 0$$

$$\gamma^2 = (R' + j\omega L')(G' + j\omega C')$$

$$\gamma = \alpha + j\beta \qquad \alpha = Re\{\gamma\} \qquad \beta = Im\{\gamma\}$$

The solutions are of the form,

$$V(z) = V_o^+ e^{-\gamma z} + V_o^- e^{\gamma z}$$

$$I(z) = I_o^+ e^{-\gamma z} + I_o^- e^{\gamma z}$$

$$I(z) = \frac{1}{Z_o}\left(V_o^+ e^{-\gamma z} - V_o^- e^{\gamma z}\right)$$

Also, from the line equations, we write

$$I(z) = \frac{\gamma}{R' + j\omega L'}\left(V_o^+ e^{-\gamma z} - V_o^- e^{\gamma z}\right)$$

$$I_o^+ = \frac{\gamma V_o^+}{R' + j\omega L'} \qquad I_o^- = \frac{-\gamma V_o^-}{R' + j\omega L'}$$

Or,

$$I_o^+ = \frac{(G' + j\omega C')V_o^+}{\gamma} \qquad I_o^- = \frac{-(G' + j\omega C')V_o^-}{\gamma}$$

We can define "line impedance" as,

$$Z_o = \frac{V_o^+}{I_o^+} = -\frac{V_o^-}{I_o^-}$$

$$Z_o = \frac{R' + j\omega L'}{\gamma} = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}}$$

Characteristic Impedance (Ω)

$$Z_o = R_o + jX_o$$

Lossless Line

$$G' = 0$$

$$R' = 0$$

$$\gamma = \sqrt{(j\omega L')(j\omega C')}$$

$$\alpha = 0$$

$$\beta = \omega\sqrt{L'C'} = \omega\sqrt{\mu\varepsilon}$$

$$Z_o = \sqrt{\frac{L'}{C'}}$$

$$v = c/\sqrt{\varepsilon_r}$$

$$\lambda = 2\pi/\beta$$

Distortionless Line

Attenuation is frequency-independent
Phase is linearly dependent of frequency

$$\frac{R'}{L'} = \frac{G'}{C'}$$

$$\gamma = \sqrt{R'G'}\left(1 + \frac{j\omega C'}{G'}\right)$$

$$\alpha = \sqrt{R'G'}$$

$$\beta = \omega\sqrt{L'C'} = \omega\sqrt{\mu\varepsilon}$$

$$Z_o = \sqrt{\frac{R'}{G'}} = \sqrt{\frac{L'}{C'}}$$

Examples

1) Design a 50-Ohm air line at 700 MHz with phase constant of 20 rad/m.

$$\frac{\beta}{Z_o} = \omega C' = \frac{20}{50} \quad \Rightarrow \quad C' = \frac{20}{50\,\omega} = 90.9\ pF/m$$

$$L' = Z_o{}^2\, C' = 227\ nH/m$$

2) Find the circuit elements of a 60-Ohm distortionless line with attenuation constant of 20 mNp/m and wave speed of 0.6c.

$$\alpha Z_o = \sqrt{R'G'}\sqrt{\frac{R'}{G'}} = R' \quad \Rightarrow \quad R' = 1.2\ \Omega/m$$

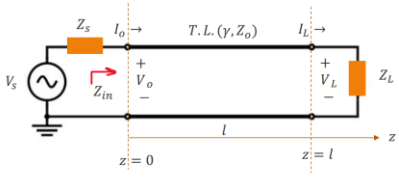
$$\frac{Z_o}{v} = \frac{\sqrt{L'C'}}{1/\sqrt{L'C'}} = L' \quad \Rightarrow \quad L' = 333\ nH/m$$

$$\frac{\alpha}{Z_o} = \frac{\sqrt{R'G'}}{\sqrt{\frac{R'}{G'}}} = G' \quad \Rightarrow \quad G' = 0.33\ mS/m$$

$$C' = 92.6\ pF/m$$

Analysis of Transmission Line Circuits

Consider the following general circuit with the following axis set-up



$$V(z) = V_o^+ e^{-\gamma z} + V_o^- e^{\gamma z}$$

$$I(z) = \frac{1}{Z_o}\left(V_o^+ e^{-\gamma z} - V_o^- e^{\gamma z}\right)$$

To find the two unknowns (V_o^+ and V_o^-), boundary conditions are needed.

1) From the input (source) side.
2) From the output (load) side.

If input voltage and current are known, then

$$V(z=0) = V_o^+ + V_o^- = V_o$$

$$I(z=0) = \frac{1}{Z_o}\left(V_o^+ - V_o^-\right) = I_o$$

Solve for the unknowns

$$V_o^+ = \frac{1}{2}(V_o + Z_o I_o)$$

$$V_o^- = \frac{1}{2}(V_o - Z_o I_o)$$

Note that: $\quad V_o = V_s\,\frac{Z_{in}}{Z_{in} + Z_s} \qquad I_o = \frac{V_s}{Z_{in} + Z_s}$

If output voltage and current are known, then

$$V(z=l) = V_o^+ e^{-\gamma l} + V_o^- e^{\gamma l} = V_L$$

$$I(z=l) = \frac{1}{Z_o}\left(V_o^+ e^{-\gamma l} - V_o^- e^{\gamma l}\right) = I_L$$

Solve for the unknowns

$$V_o^+ = \frac{1}{2}(V_L + Z_o I_L)\, e^{\gamma l}$$

$$V_o^- = \frac{1}{2}(V_L - Z_o I_L)\, e^{-\gamma l}$$

To evaluate the input impedance, Z_{in}

$$Z_{in} = Z(0) = \frac{V(0)}{I(0)} = \frac{V_o}{I_o} = \frac{V_o^+ + V_o^-}{\frac{1}{Z_o}\left(V_o^+ - V_o^-\right)}$$

$$Z_{in} = Z_o\left[\frac{Z_L + Z_o\tanh(\gamma l)}{Z_o + Z_L\tanh(\gamma l)}\right]$$

Lossy line

$$Z_{in} = Z_o\left[\frac{Z_L + jZ_o\tan(\beta l)}{Z_o + jZ_L\tan(\beta l)}\right]$$

Lossless line

A 2-m-long transmission line operating at 10^6 rad/s has the following properties α=8 dB/m, β=1 rad/m and Z_o=60+j40 Ohms. The line is connected to a source with V_s=10∠0 V and Z_s=40 Ohms. The connected load is Z_L=20+j50 Ohms. Find Z_{in}, I_o and $I(L/2)$.

$$\alpha = \frac{8}{8.686} = 0.921\ Np/m \quad \Rightarrow \quad \gamma = 0.921 + j \quad (1/m)$$

$$\gamma L = 2(0.921 + j) \qquad \tanh\gamma L = 1.033 - j0.03929 \quad \Rightarrow \quad Z_{in} = 60.25 + j\,38.79\ Ohms$$

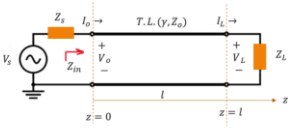
$$I_o = \frac{V_s}{Z_s + Z_{in}} = \frac{10\angle 0}{40 + 60.25 + j\,38.79} = 93.03\angle -21.75° \ (mA)$$

$$V_o = I_o Z_{in} = 6.667\angle 11.62° \ (V)$$

$$V_o^+ = \frac{1}{2}(V_o + Z_o I_o) = 6.687\angle 12.08° \ (V) \qquad I(z = L/2) = \frac{V_o^+}{Z_o}e^{-\gamma L/2} - \frac{V_o^-}{Z_o}e^{\gamma L/2}$$

$$V_o^- = \frac{1}{2}(V_o - Z_o I_o) = 0.0518\angle 260° \ (V) \qquad I(z = L/2) = 35.1\angle 281° \ (mA)$$

Consider the following general circuit with the following axis set-up



To evaluate the voltage reflection $\quad \Gamma(z) = \frac{V_o^- e^{\gamma z}}{V_o^+ e^{-\gamma z}} \qquad \Gamma(0) = \frac{V_o^-}{V_o^+}$

At load $(z = l)$ $\quad \Gamma_L = \frac{Z_L - Z_o}{Z_L + Z_o}$ $\quad$ For a "matched" load $(Z_L = Z_o)$ $\quad \Gamma_L = 0$

At any location $\quad \Gamma(z) = \Gamma_L\, e^{-2\gamma z}$

In general, loads are complex $\quad \Gamma = |\Gamma|\, e^{j\theta_\Gamma}$

Example

Find the reflection coefficient at the load on a 100-Ohm TL at 100 MHz. The load is a 50-Ohm resistor and a 10 pF capacitor in series.

$$Z_L = R_L + 1/j\omega C_L = 50 - j/(2\pi\times 100\,M)(10\,p)$$

$$\Gamma = \frac{50 - j/(2\pi\times 100\,M)(10\,p) - 100}{50 - j/(2\pi\times 100\,M)(10\,p) + 100} = -0.76\, e^{j(119.3°)} = 0.76\angle -60.7°$$

What happens if the resistor is removed (purely reactive load)?

$$\Gamma = \frac{-j/(2\pi\times 100\,M)(10\,p) - 100}{-j/(2\pi\times 100\,M)(10\,p) + 100} = -\,e^{j(115.6°)}$$

For any purely reactive load $\quad |\Gamma| = 1 \qquad \theta_\Gamma = \tan^{-1}(X_L/Z_o)$

Reflection Analysis

Using the following relation, $\quad V_o^- = \Gamma(0)\, V_o^+$

We write for a lossless line, $\quad V(z) = V_o^+\left(e^{-j\beta z} + \Gamma e^{j\beta z}\right) \qquad I(z) = \frac{V_o^+}{Z_o}\left(e^{-j\beta z} - \Gamma e^{j\beta z}\right)$

$$|V(z)| = \sqrt{\left[V_o^+\left(e^{-j\beta z} + \Gamma e^{j\beta z}\right)\right]\cdot\left[V_o^+\left(e^{-j\beta z} + \Gamma e^{j\beta z}\right)\right]^*}$$

$$|V(z)| = |V_o^+|\sqrt{1 + |\Gamma|^2 + |\Gamma|e^{j(2\beta z + \theta_\Gamma)} + |\Gamma|e^{-j(2\beta z + \theta_\Gamma)}}$$

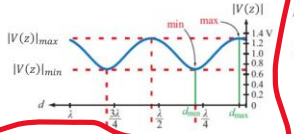
$$|V(z)| = |V_o^+|\sqrt{1 + |\Gamma|^2 + 2|\Gamma|\cos(2\beta z + \theta_\Gamma)}$$

$$|V(z)|_{max} = |V_o^+|\,(1 + |\Gamma|) \qquad |V(z)|_{min} = |V_o^+|\,(1 - |\Gamma|)$$

$$S = \frac{|V(z)|_{max}}{|V(z)|_{min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

Voltage Standing Wave Ratio (VSWR)

A measure of how good is the load matching. VSWR=1 perfect match

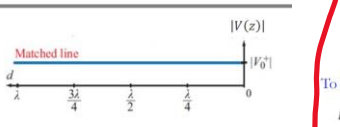


Special Cases

1) Matched Line

$Z_L = Z_o$

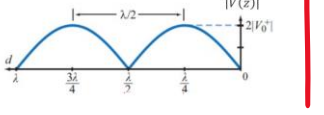
$\Gamma = 0$



2) Shorted Line

$Z_L = 0$

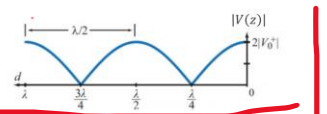
$\Gamma = -1$



3) Open-circuited Line

$Z_L = \infty$

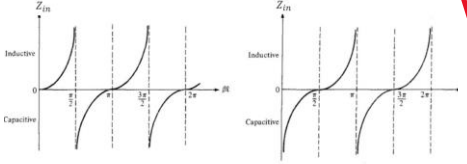
$\Gamma = 1$



Special Cases

Input impedance versus line length for the two special cases.

$$Z_{in} = Z_o\left[\frac{Z_L + jZ_o\tan(\beta l)}{Z_o + jZ_L\tan(\beta l)}\right]$$



Shorted Line $\qquad$ Open-circuited Line

Power Analysis

Starting from the equations, $\quad V(z) = V_o^+\left(e^{-j\beta z} + \Gamma e^{j\beta z}\right) \qquad I(z) = \frac{V_o^+}{Z_o}\left(e^{-j\beta z} - \Gamma e^{j\beta z}\right)$

In general, $\quad V_o^+ = |V_o^+|e^{j\phi}$

We need instantaneous expressions.

$$v(z,t) = |V_o^+|\left[\cos(\omega t + \beta z + \phi) + |\Gamma|\cos(\omega t - \beta z + \phi + \theta_\Gamma)\right]$$

$$i(z,t) = \frac{|V_o^+|}{Z_o}\left[\cos(\omega t + \beta z + \phi) - |\Gamma|\cos(\omega t - \beta z + \phi + \theta_\Gamma)\right]$$

$$P(z,t) = \frac{|V_o^+|^2}{Z_o}\left[\cos^2(\omega t + \beta z + \phi) - |\Gamma|^2\cos^2(\omega t + \beta z + \phi + \theta_\Gamma)\right]$$

Incident power $\qquad$ Reflected power

$$P^i(z,t) = \frac{|V_o^+|^2}{2Z_o}\left[1 + \cos(2\omega t + 2\beta z + 2\phi)\right] \qquad P^r(z,t) = -|\Gamma|^2\frac{|V_o^+|^2}{2Z_o}\left[1 + \cos(2\omega t + 2\beta z + 2\phi + 2\theta_\Gamma)\right]$$

DC $\quad$ AC $\qquad$ DC $\quad$ AC

$$P^i{}_{ave} = \frac{|V_o^+|^2}{2Z_o} \qquad P^r{}_{ave} = -|\Gamma|^2\frac{|V_o^+|^2}{2Z_o} = -|\Gamma|^2\, P^i{}_{ave}$$

$$P_{ave} = \frac{|V_o^+|^2}{2Z_o}(1 - |\Gamma|^2)$$

Independent of location along the line

Examples

(1) A 30-m-long lossless transmission line with $Z_o = 50$ Ohms at 2 MHz is terminated by a load with $Z_L = 60 + j40$ Ohms. The wave velocity on the line is 0.6c. Find Γ_L, $VSWR$ and Z_{in}.

$$\Gamma_L = \frac{Z_L - Z_o}{Z_L + Z_o} = \frac{60 + j40 - 50}{60 + j40 + 50} = 0.3523\angle 56°$$

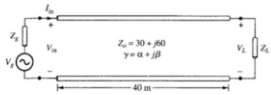
$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.3523}{1 - 0.3523} = 2.088$$

$$\beta = \frac{\omega}{v} \quad \Rightarrow \quad \beta l = \frac{\omega l}{v} = \frac{2\times 10^6 \times 30}{0.6\times 3\times 10^8} = \frac{2\pi}{3}$$

$$Z_{in} = Z_o\left[\frac{Z_L + jZ_o\tan(\beta l)}{Z_o + jZ_L\tan(\beta l)}\right] = 50\,\frac{(60 + j40) + j50\tan 120}{50 + j(60 + j40)\tan 120} = 24\angle 3.22°\ \Omega$$

Examples

(2) A 40-m-long transmission line with $Z_o = 30 + j60$ Ohms. The load is matched to the line. Given $Z_g = 30 + j60$ Ohms, $V_g = 15\angle 0\ V$(rms) and $V_L = 5\angle -48°\ V$(rms), find Z_{in}, I_{in}, V_{in} and γ.



$$Z_L = Z_o$$

$$Z_{in} = Z_o\left[\frac{Z_L + Z_o\tanh(\gamma l)}{Z_o + Z_L\tanh(\gamma l)}\right] = Z_o = 30 + j60\ \Omega$$

$$I_{in} = \frac{V_g}{Z_{in} + Z_g} = \frac{15\angle 0}{2(30 + j60)} = 0.112\angle -63.4°\ (A)$$

$$V_{in} = \frac{V_g}{2} = 7.5\angle 0\ (V\ rms)$$

$$V_o^+ = \frac{1}{2}(V_{in} + I_{in}Z_o) = \frac{1}{2}(V_L + Z_o I_L)\, e^{\gamma l} \quad \Rightarrow \quad e^{\gamma l} = \frac{V_{in} + I_{in}Z_o}{V_L + Z_o I_L}$$

$$\gamma = 0.0101 + 0.2094j \quad (1/m)$$

Metallic Waveguides

Without going into complicated details, we can construct the required solutions by inspection.

1) TM Modes ($H_z = 0$)

$$E_z = (A\cos\beta_x x + B\sin\beta_x x)\cdot(C\cos\beta_y y + D\sin\beta_y y)\cdot e^{-j\beta_z z}$$

Apply boundary conditions, tangential fields=0 at $x = 0, a$ and $y = 0, b$.

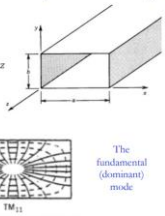
$$E_z = E_o\sin\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

$$m = 1, 2, \cdots \qquad n = 1, 2, \cdots$$

$$\beta_x = \frac{m\pi}{a} \qquad \beta_y = \frac{n\pi}{a}$$

Mode number is TM_{mn}

Modes TM_{00} and TM_{01} and TM_{10} are not possible ($E_z = 0$)



The fundamental (dominant) mode

To find other field components, use Maxwell's equations.

$$E_z = E_o\sin\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

$$E_x = \frac{-j\beta}{\beta_{xy}^2}\frac{m\pi}{a}E_o\cos\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

$$E_y = \frac{-j\beta}{\beta_{xy}^2}\frac{n\pi}{b}E_o\sin\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

$$H_x = \frac{j\omega\varepsilon}{\beta_{xy}^2}\frac{n\pi}{b}E_o\sin\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

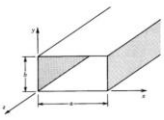
$$H_y = \frac{-j\omega\varepsilon}{\beta_{xy}^2}\frac{m\pi}{a}E_o\cos\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

$$H_z = 0$$

$$\beta_{xy}^2 = \beta_x^2 + \beta_y^2$$

$$f_{mn} = \frac{1}{2\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

Cut-off (resonance) frequency



$$\beta = \sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

Should be real for mode propagation

$$Z_{mn} = \eta\sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}$$

Without going into complicated details, we can construct the required solutions by inspection.

2) TE Modes ($E_z = 0$)

$$H_z = H_o\cos\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

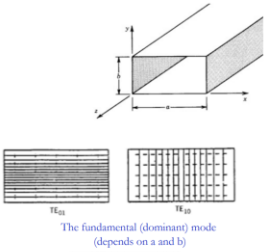
$$m = 0, 1, 2, \cdots \qquad n = 0, 1, 2, \cdots$$

$$\beta_x = \frac{m\pi}{a} \qquad \beta_y = \frac{n\pi}{a}$$

Mode number is TE_{mn}

Mode TE_{00} is not possible

But modes TE_{01} and TE_{10} are valid



The fundamental (dominant) mode (depends on a and b)

To find other field components, use Maxwell's equations.

$$H_z = H_o \cos\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

$$H_x = \frac{j\beta}{\beta_{xy}^2}\frac{m\pi}{a}H_o \sin\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

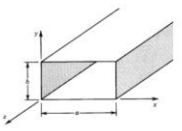
$$H_y = \frac{j\beta}{\beta_{xy}^2}\frac{n\pi}{b}H_o \cos\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

$$E_x = \frac{j\omega\mu}{\beta_{xy}^2}\frac{n\pi}{b}H_o \cos\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

$$E_y = \frac{-j\omega\mu}{\beta_{xy}^2}\frac{m\pi}{a}H_o \sin\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

$$E_z = 0$$

$$\beta_{xy}^2 = \beta_x^2 + \beta_y^2$$



$$\beta = \sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

Should be real for mode propagation

$$f_{mn} = \frac{1}{2\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

Cut-off (resonance) frequency

$$Z_{mn} = \frac{\eta}{\sqrt{1-\left(\frac{f_{mn}}{f}\right)^2}}$$

Examples

1) Consider a RWG with dimensions 2.5 cm × 1 cm, filled with a lossless material of ε_r=4. Find the cutoff frequencies of all allowed modes up to 15 GHz. Which is the dominant mode?

$$f_{mn} = \frac{c}{4}\sqrt{\left(\frac{m}{0.025}\right)^2 + \left(\frac{n}{0.01}\right)^2}$$

TE_{mn}	f_{mn} (GHz)	TM_{mn}	f_{mn} (GHz)
10	3	10	
01	7.5	01	
11	8.078	11	8.078
02	15	12	15.3
03	22.5	21	9.6
20	6	31	11.72
30	9	41	14.14
40	12		
50	15		
60	18		

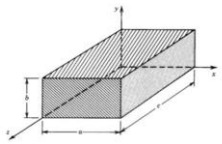
11 TE modes
4 TM modes

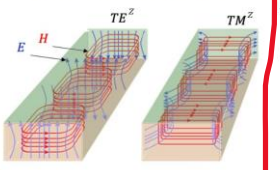
Dominant mode: TE_{10}

Single mode operation between 3-6 GHz

Metallic Cavities

The concept of wave confinement here is based on perfect reflection over the metallic surfaces. The wave is confined to the space filling the cavity (low-loss).





For a given cavity dimensions (a, b, c), an infinite number of different wave profiles (modes) are possible, each with its own frequency, wavelength, etc.

Waves in MRC are non-TEM; classified as either TE or TM.

Wave Equation Solutions: TM Modes

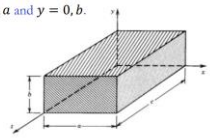
Without going into complicated details, we can construct the required solutions by inspection.

1) TM Modes ($H_z = 0$)

$$E_z = (C_1\cos\beta_x x + C_2\sin\beta_x x)\cdot(C_3\cos\beta_y y + C_4\sin\beta_y y)\cdot(C_5\cos\beta_z z + C_6\sin\beta_z z)$$

Apply boundary conditions, tangential fields=0 at $x = 0, a$ and $y = 0, b$.

$$E_z = E_o\cdot\sin\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot\cos\left(\frac{p\pi}{c}z\right)$$



$m = 1, 2, \cdots$ $\quad n = 1, 2, \cdots$ $\quad p = 0, 1, 2, \cdots$

$$\beta_x = \frac{m\pi}{a} \qquad \beta_y = \frac{n\pi}{b} \qquad \beta_z = \frac{p\pi}{c}$$

$$\beta^2 = \omega^2\varepsilon\mu = \beta_x^2 + \beta_y^2 + \beta_z^2$$

Mode number is TM_{mnp}

Modes TM_{00p}, TM_{01p}, and TM_{10p} are not possible ($E_z = 0$)

Wave Equation Solutions: TM Modes

To find other field components, use Maxwell's equations.

$$E_z = E_o\cdot\sin\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot\cos\left(\frac{p\pi}{c}z\right)$$

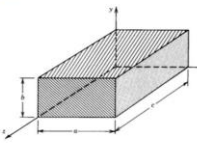
$$E_x = \frac{-1}{\beta^2}\left(\frac{m\pi}{a}\right)\left(\frac{p\pi}{c}\right)E_o\cdot\cos\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot\sin\left(\frac{p\pi}{c}z\right)$$

$$E_y = \frac{-1}{\beta^2}\left(\frac{n\pi}{b}\right)\left(\frac{p\pi}{c}\right)E_o\cdot\sin\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot\sin\left(\frac{p\pi}{c}z\right)$$

$$H_x = \frac{j\omega\varepsilon}{\beta^2}\left(\frac{n\pi}{b}\right)E_o\cdot\sin\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot\cos\left(\frac{p\pi}{c}z\right)$$

$$H_y = \frac{-j\omega\varepsilon}{\beta^2}\left(\frac{m\pi}{a}\right)E_o\cdot\cos\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot\cos\left(\frac{p\pi}{c}z\right)$$

$$H_z = 0$$



$$f_{mnp} = \frac{1}{2\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{c}\right)^2}$$

Cut-off (resonance) frequency for mode TM_{mnp}

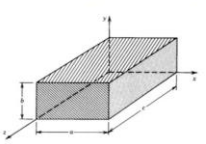
The dominant mode is the mode that will be excited first and has the lowest resonance frequency (TM_{110}).

Wave Equation Solutions: TE Modes

Without going into complicated details, we can construct the required solutions by inspection.

2) TE Modes ($E_z = 0$)

$$H_z = H_o\cdot\cos\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot\sin\left(\frac{p\pi}{c}z\right)$$



$m = 0, 1, 2, \cdots$ $\quad n = 0, 1, 2, \cdots$ $\quad p = 1, 2, \cdots$

$$\beta_x = \frac{m\pi}{a} \qquad \beta_y = \frac{n\pi}{b} \qquad \beta_z = \frac{p\pi}{c}$$

$$\beta^2 = \omega^2\varepsilon\mu = \beta_x^2 + \beta_y^2 + \beta_z^2$$

Mode number is TE_{mnp}

Modes TE_{mn0} are not possible

But modes TE_{m0p} and TE_{0np} are valid

Note: m and n cannot be both zero at the same time

Wave Equation Solutions: TE Modes

To find other field components, use Maxwell's equations.

$$H_z = H_o\cdot\cos\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot\sin\left(\frac{p\pi}{c}z\right)$$

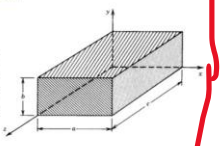
$$H_x = \frac{-1}{\beta^2}\left(\frac{m\pi}{a}\right)\left(\frac{p\pi}{c}\right)H_o\cdot\sin\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot\cos\left(\frac{p\pi}{c}z\right)$$

$$H_y = \frac{-1}{\beta^2}\left(\frac{n\pi}{b}\right)\left(\frac{p\pi}{c}\right)H_o\cdot\cos\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot\cos\left(\frac{p\pi}{c}z\right)$$

$$E_x = \frac{j\omega\mu}{\beta^2}\left(\frac{n\pi}{b}\right)H_o\cdot\cos\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot\sin\left(\frac{p\pi}{c}z\right)$$

$$E_y = \frac{-j\omega\mu}{\beta^2}\left(\frac{m\pi}{a}\right)H_o\cdot\sin\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot\sin\left(\frac{p\pi}{c}z\right)$$

$$E_z = 0$$



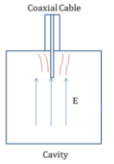
$$f_{mnp} = \frac{1}{2\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{c}\right)^2}$$

Cut-off (resonance) frequency for mode TE_{mnp}

The dominant mode is the mode that will be excited first and has the lowest resonance frequency (TE_{101} or TE_{011}).

Metallic Rectangular Cavity

A cavity can be excited/probed using a number of methods. A coaxial cable tip can be inserted through a small hole, or an aperture can be made in the wall. The probe location has to match the profile of the mode being sensed/excited.



The quality factor for any resonator is defined as,

$$Q = 2\pi\frac{Time - average\ Energy\ stored}{Energy\ loss\ per\ cycle}$$

For the dominant mode of a metallic rectangular cavity,

$$Q_{TE_{101}} = \frac{abc(a^2+c^2)}{\delta[2b(a^3+c^3)+ac(a^2+c^2)]}$$

where the skin depth of the cavity metallic walls

$$\delta = \frac{1}{\sqrt{\pi f_{101}\mu_o\sigma_c}}$$

σ_c is the metal conductivity

Examples

(1) Consider an MRC with dimensions 5 cm × 4 cm × 10 cm, filled with air and made of copper ($\sigma_c = 5.8\times10^7$ S/m). Find the cutoff frequencies of all modes up to 5 GHz. Find the quality factor of the dominant mode?

$$f_{mnp} = \frac{c}{2}\sqrt{\left(\frac{m}{0.05}\right)^2 + \left(\frac{n}{0.04}\right)^2 + \left(\frac{p}{0.1}\right)^2}$$

TE_{mnp}	f_{mnp} (GHz)	TM_{mnp}	f_{mnp} (GHz)
011	4.04	110	4.80
101	3.35	111	5.03
111	5.03		
102	4.24		
012	4.80		

4 TE modes
1 TM mode

Dominant mode: TE_{101}

Single mode operation between 3.35-4.04 GHz

$Q_{TE101} = 14{,}358$

Examples

(2) Consider a dielectric-filled rectangular cavity with dimensions a=5 cm, b=3 cm and c=3 cm. One of the TM modes that are excited inside this cavity has x-component magnetic field given by

$$H_x(x,y,z,t) = 36\sin(40\pi x)\cdot\cos(100\pi y)\cdot\sin(6.4\pi\times10^9\,t) \quad (mA/m)$$

Find (a) the number of this mode, (b) the dielectric constant of the cavity filling material and (c) all the TE modes that are excited before the given mode.

$$H_x = \frac{j\omega\varepsilon}{\beta^2}\left(\frac{n\pi}{b}\right)E_o\cdot\sin\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot\cos\left(\frac{p\pi}{c}z\right)$$

$$\beta_x = \frac{m\pi}{a} = 40\pi \rightarrow m = 2$$

$$\beta_y = \frac{n\pi}{b} = 100\pi \rightarrow n = 3$$

$$\beta_z = \frac{p\pi}{b} = 0 \rightarrow p = 0$$

TM_{230}

$$f_{230} = \frac{c}{2\sqrt{\varepsilon_r}}\sqrt{\left(\frac{2}{0.05}\right)^2 + \left(\frac{3}{0.03}\right)^2 + \left(\frac{0}{0.03}\right)^2} = 3.2\times10^9\ (Hz)$$

$\varepsilon_r = 25$

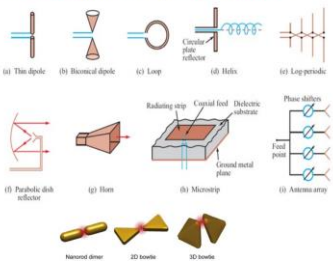
TE Modes

f_{110} = ✗	f_{101} = ✓ 1.166e+9	f_{011} = ✓ 1.414e+9
f_{111} = ✓ 1.536e+9	f_{201} = ✓ 1.562e+9	f_{120} = ✗ 2.088e+9
f_{211} = ✓ 1.855e+9	f_{210} = ✗	f_{021} = ✓ 2.236e+9
f_{220} = ✗ 2.332e+9	f_{221} = ✓ 2.538e+9	f_{121} = ✓ 2.315e+9

Antennas

Antennas are metallic structures that are designed specifically to radiate (receive) EM waves efficiently, within a specific frequency band, with specific characteristics.

They come in many different types and shapes, each with



- Basic shapes.
- Principle of radiation.
- Field profiles (modes).
- Frequency range.
- Radiation patterns.
- Advantages.
- Limitations.

Antennas: Introduction

Antennas are characterized based on several parameters. The application of a certain type of antenna in a certain application is based on the nature of the problem.

Basic parameters include:

1) Resonance frequency: the frequency at which the antenna efficiently converts incoming power to EM waves, or have best reception.
2) Bandwidth: the range of frequencies where the operation of an antenna is acceptable.
3) Polarization: linear, circular or elliptical.
4) Radiation pattern: the space map of antenna radiation.
5) Efficiency: the conversion ratio of supplied power to radiated power.
6) Gain: the ratio of radiated power in a certain direction to a reference antenna.
7) Directivity: a measure of the focus of a radiation beam.

The Hertzian Dipole

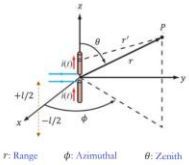
This is a basic antenna element, a very short wire carrying current, or a differential antenna (a point source).

Basic assumptions:

1) Infinitesimal length compared to distances involved (the wavelength)
$l \ll \lambda \quad l < \lambda/50$

2) The current along the wire becomes constant.
$I(z) = I_o\cos\omega t \approx I_o$

3) The radiated waves are spherical.



r: Range $\quad$ ϕ: Azimuthal $\quad$ θ: Zenith

Spherical coordinates simplify the work

$$\vec{E} = E_r\vec{a}_r + E_\theta\vec{a}_\theta + E_\phi\vec{a}_\phi$$

$$\vec{H} = H_r\vec{a}_r + H_\theta\vec{a}_\theta + H_\phi\vec{a}_\phi$$

Because the current flows in the z-direction, the resulting fields have the following components

E_r, E_θ, H_ϕ

$E_\phi = H_\theta = H_r = 0$

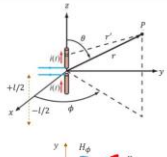
The Hertzian Dipole

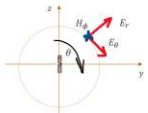
The existing field are as follows.

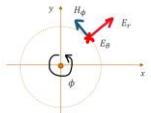
$$E_r = \frac{2I_o\,l\,\beta^2}{4\pi}\eta_o\left[\frac{1}{(\beta r)^2} - \frac{j}{(\beta r)^3}\right]\cos\theta\, e^{-j\beta r}$$

$$E_\theta = \frac{I_o\,l\,\beta^2}{4\pi}\eta_o\left[\frac{j}{\beta r} + \frac{1}{(\beta r)^2} - \frac{j}{(\beta r)^3}\right]\sin\theta\, e^{-j\beta r}$$

$$H_\phi = \frac{I_o\,l\,\beta^2}{4\pi}\left[\frac{j}{\beta r} + \frac{1}{(\beta r)^2}\right]\sin\theta\, e^{-j\beta r}$$



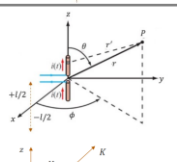




Far-field Approximation.

r very large $\quad \beta r \gg 1$

$\frac{2\pi}{\lambda}r \gg 1 \rightarrow r \gg \frac{\lambda}{2\pi}$ Far-field Zone



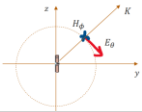
$$E_r \approx 0$$

$$E_\theta = \frac{jI_o\,l\,\beta}{4\pi}\eta_o\sin\theta\left[\frac{e^{-j\beta r}}{r}\right]$$

$$H_\phi = \frac{jI_o\,l\,\beta}{4\pi}\sin\theta\left[\frac{e^{-j\beta r}}{r}\right]$$

$$H_\phi = E_\theta/\eta_o$$

Plane-wave behavior

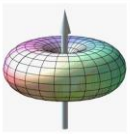


Time-average Power density

$$\vec{S}_{ave} = \frac{1}{2}Re(\vec{E}\times\vec{H}^*)$$

$$\vec{S}_{ave} = \frac{\eta_o\beta^2 I_o^2 l^2}{32\pi^2 r^2}(\sin\theta)^2\,\vec{a}_r$$

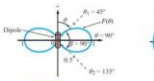
$$\vec{S}_{ave} = \frac{\eta_o I_o^2}{8r^2}\left(\frac{l}{\lambda}\right)^2(\sin\theta)^2\,\vec{a}_r$$



$$\vec{S}_{ave} = S_o(r)\,(\sin\theta)^2\,\vec{a}_r$$

$$S_o(r) = \frac{\eta_o I_o^2}{8r^2}\left(\frac{l}{\lambda}\right)^2$$

$$S_{ave_max} = S_o(r)$$



Elevation plane $\phi = 0$

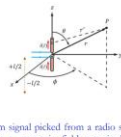


Azimuthal plane $\theta = \frac{\pi}{2}$

Radiated Power

$$P_{rad} = \oint_S \vec{S}_{ave}\cdot\vec{a}_S$$

$$P_{rad} = \int_0^{2\pi}\int_0^{\pi}\frac{\eta_o I_o^2}{8r^2}\left(\frac{l}{\lambda}\right)^2(\sin\theta)^2\, r^2\sin\theta\, d\theta d\phi$$



$$P_{rad} = \frac{\pi\eta_o I_o^2}{3}\left(\frac{l}{\lambda}\right)^2$$

Example

The maximum signal picked from a radio station 2 km away has a magnetic field magnitude of 5 $\mu A/m$. Find the station radiated power. $l/\lambda = 1/25$.

$$|H_\phi| = \frac{I_o\,l\,\beta}{4\pi r}\sin\theta = 5\mu \rightarrow I_o = 0.5\,A$$

$$P_{rad} = \frac{\pi\eta_o I_o^2}{3}\left(\frac{l}{\lambda}\right)^2 = 157.7\,mW$$

$$R_{rad} = 80\pi^2\left(\frac{l}{\lambda}\right)^2 = 1.26\ \Omega$$

Radiation Resistance

$$P_{rad} = I_{rms}^2 R_{rad}$$

$$P_{rad} = \frac{1}{2}I_o^2 R_{rad}$$

$$R_{rad} = 80\pi^2\left(\frac{l}{\lambda}\right)^2 \approx 1\ \Omega$$

Efficiency

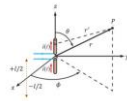
$$\eta = \frac{P_{rad}}{P_{tot}} = \frac{P_{rad}}{P_{rad}+P_{loss}}$$

Gain

$$G = \frac{S_o}{P_{tot}/4\pi r^2}$$

Directivity

$$D = \frac{S_o}{P_{rad}/4\pi r^2} \approx 1.5$$



Example: A 1-m dipole is excited by a 5-MHz current of amplitude 5 A. Find the broadside power density at 2 km.

Check if it's a Hertzian dipole. $\lambda = \frac{v}{f} = 3\times10^8/5\times10^6 = 60\ m \qquad \frac{l}{\lambda} = \frac{1}{60}$

$$S_{ave} = \frac{\eta_o I_o^2}{8(2000)^2}\left(\frac{l}{\lambda}\right)^2(\sin 90°)^2 = 8.2\times10^{-8}\ W/m^2$$

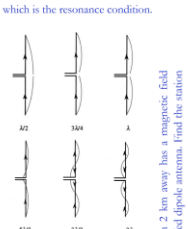
The Dipole Wire Antenna

The dipole antenna length is $\lambda/2$, or any multiple, which is the resonance condition.

The analysis follow the same steps as the Hertzian dipole with one very important difference: the current distribution along the antenna is not constant.

The current along the wire is

$$I(z) = I_o\cos\beta z = I_o\cos\frac{2\pi}{\lambda}z$$



Far-field Approximation.

$$E_\theta = j60\,I_o\frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}\left[\frac{e^{-j\beta r}}{r}\right]$$

$$H_\phi = \frac{E_\theta}{\eta_o} = \frac{jI_o}{2\pi}\frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}\left[\frac{e^{-j\beta r}}{r}\right]$$

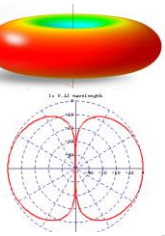
The Dipole Wire Antenna

Power density

$$\vec{S}_{ave} = \frac{\eta_o I_o^2}{8\pi^2 r^2}\left(\frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}\right)^2\vec{a}_r$$

$$\vec{S}_{ave} = S_o(r)\left(\frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}\right)^2\vec{a}_r$$

$$S_o(r) = \frac{15\,I_o^2}{\pi\,r^2}$$



Radiated Power

$$P_{rad} = \oint_S \vec{S}_{ave}\cdot\vec{a}_S \approx 36.56\,I_o^2$$

$$R_{rad} \approx 73\ \Omega$$

$$D \approx 1.64$$

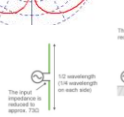
The mono-pole wire antenna

$l = \lambda/4$

$P_{rad} \approx 18.28\,I_o^2$

$R_{rad} \approx 36.5\ \Omega$

$0 \le \theta \le \frac{\pi}{2}$



The maximum signal picked from a radio station 2 km away has a magnetic field magnitude of 5 $\mu A/m$. The station uses a center-fed dipole antenna. Find the station radiated power.

$$|H_\phi| = \frac{I_o}{2\pi r} = 5\mu \rightarrow I_o = 0.063\,A$$

$$P_{rad} = (36.57)(0.063)^2 = 144\ mW$$